

Selective formation of light olefins from CO₂ hydrogenation over Fe–Zn–K catalysts



Jianli Zhang*, Shipeng Lu, Xiaojuan Su, Subing Fan, Qingxiang Ma, Tiansheng Zhao*

State Key Laboratory Cultivation Base of Natural Gas Conversion, College of Chemical Engineering, Ningxia University, Yinchuan 750021, People's Republic of China

ARTICLE INFO

Article history:

Received 29 January 2015

Received in revised form 22 April 2015

Accepted 28 May 2015

Available online 13 June 2015

Keywords:

Microwave-aided hydrothermal procedure

Fe–Zn–K catalyst

CO₂ hydrogenation

Light olefins

K promotion

ABSTRACT

Fe–Zn–K catalysts were prepared in varied Fe/Zn molar ratios using microwave-aided hydrothermal procedure followed by K modification with impregnation method and applied to CO₂ hydrogenation reaction via Fischer–Tropsch synthesis for the selective production of C₂–C₄ olefins. Results showed that the prepared catalysts had uniform particles within about 100 nm. Addition of zinc to iron matrix formed ZnFe₂O₄ spinel phase and ZnO phase, and caused increase of the surface areas, enhanced the interaction between iron and zinc and altered the reduction and CO₂ adsorption behaviors. The catalysts displayed high activity for the CO₂ conversion and significant improvement in the product distribution. The proper interactions between Fe and Zn proved to be advantageous to suppress the production of C₅⁺ hydrocarbons and promote the production of C₂–C₄ olefin. At set reaction conditions of H₂/CO₂ of 3, GHSV of 1000 h^{−1}, 320 °C, and 0.5 MPa, the 1Fe–1Zn–K catalyst with H₂/CO reduction showed the best performance with the CO₂ conversion of 51.03%. The selectivity of C₂–C₄ olefins in overall hydrocarbons and the ratio of olefin to paraffin in the C₂–C₄ fraction reached 53.58% and 6.86, respectively.

©2015 Elsevier Ltd. All rights reserved.

1. Introduction

The utilization of CO₂ as carbon feedstock to obtain highly added-value chemicals is of importance [1,2]. Among various transformation routes, the chemical conversion of CO₂ via hydrogenation has been considered as one of the most promising processes to produce chemicals such as methanol, higher alcohols and light olefins over modified Fischer–Tropsch synthesis (FTS) [1–5]. Direct synthesis of lower olefins from CO₂ hydrogenation is desired from the viewpoint of converting CO₂ to the building-block chemicals and has progressed in recent years [3,4]. To raise the selectivity toward light olefin products, more effective catalysts need to be discovered considering the thermodynamic inertness of CO₂ and the poor product distribution via FTS.

Traditional FTS catalysts such as iron-based catalysts has been used in CO₂ hydrogenation [1–7] due to high activity for both the water gas shift (WGS) and the reverse water-gas shift (RWGS) reactions during F–T synthesis [4,8,9]. CO₂ hydrogenation on iron catalysts has manifested to proceed via a two-step process with initial conversion of CO₂ to CO via the RWGS reaction followed by the conversion of CO to hydrocarbons via F–T synthesis [4,8]. Iron

catalysts are employed in the hydrocarbon synthesis because of high selectivity toward the formation of primary olefins [4]. And CO₂ hydrogenation over iron catalysts can produce more light hydrocarbons compared with CO hydrogenation [6]. Therefore, iron based catalysts have been paid much attention in CO₂ hydrogenation [8]. Preparation methods and promoter addition show impacts on the catalytic activity. It is well recognized that potassium addition in iron catalysts enhances the production of olefinic hydrocarbons in CO hydrogenation [6,10,11]. This alkali promotion is also indispensable to raise light olefin selectivity in CO₂ hydrogenation and has been discussed in detail [3,7,12]. Potassium modified Fe/ZrO₂ catalysts prepared by impregnation method was reported for CO₂ hydrogenation with 42% of CO₂ conversion and 46% fraction of C₂–C₄ olefins in all hydrocarbon products [3]. Further, activity improvement for CO₂ hydrogenation necessitates promoter addition in iron catalysts to enhance the CO₂ adsorption while reduce the secondary hydrogenation ability and suppress heavy products. Manganese and copper, which function as both structural and electronic promoters of modified iron catalysts for CO₂ hydrogenation, have been studied and showed improved activity and selectivity toward light olefins [13,14]. Moreover, it is noteworthy that zinc promoter has been demonstrated to accelerate the FTS reaction and increase stability of the catalysts [15], to enhance the light olefin selectivity and improve CO₂ adsorption [15,16], to promote the hydrogen

* Corresponding authors. Tel.: +86 951 2062323; fax: +86 951 2062323.

E-mail addresses: zhangjl@nxu.edu.cn (J. Zhang), zhaots@nxu.edu.cn (T. Zhao).

dissociation [17] and be highly active in both the WGS and the RWGS reactions [18,19]. Although the promotion effect of zinc on FTS and methanol synthesis catalysts have been examined [15], the effect on the catalytic hydrogenation of CO₂ to hydrocarbons has not been addressed well.

In this work, a series of potassium promoted Fe–Zn catalysts with varied Fe/Zn molar ratios were prepared by microwave-aided hydrothermal procedure followed by incipient wetness impregnation. The catalysts exhibited excellent activity for CO₂ hydrogenation and high selectivity toward C₂–C₄ olefins owing to the promotion effects of zinc on iron catalysts. The relationship between the structure and the surface properties of the catalysts was discussed by means of characterization of physicochemical properties.

2. Experimental

2.1. Catalyst preparation

Fe–Zn catalysts were prepared using Fe(NO₃)₃·9H₂O, Zn(NO₃)₂·6H₂O as Fe, Zn sources, respectively, and urea as precipitator. The Fe/Zn molar ratios for series of preparation samples were 3:1, 1:1, 1:3 and 1:5 and the (Fe+Zn)/urea molar ratio was 1:2 for all the samples. First, desired amounts of Fe(NO₃)₃·9H₂O, Zn(NO₃)₂·6H₂O and urea were dissolved and mixed in desired amount of deionized water. Then, the mixed solution was put into a 100 mL Teflon-lined stainless steel autoclave, being kept at microwave conditions of 2450 MHz, 400 W and 1.6 MPa for 90 min. The pH value of the final mixture was about 11. Next, the products were filtered, washed, dried at 120 °C overnight and calcined in the air at 450 °C for 4 h. The obtained powder samples were further impregnated with the aqueous solution of potassium carbonate in the Fe/K molar ratio of 10:1, and dried at 120 °C for 8 h. Finally, all the catalyst samples were pressed into pellets (20 MPa), crushed and sieved to obtain 20–40 mesh for characterization and activity test. The obtained catalyst samples were named as Fe₂O₃-K, 3Fe-1Zn-K, 1Fe-1Zn-K, 1Fe-3Zn-K and 1Fe-5Zn-K, respectively. For performance comparison, other samples labeled as 1Fe-1Mn-K, Fe₂O₃ and ZnO without K promotion were prepared following the same procedure as mentioned above.

2.2. Catalyst characterization

The BET surface areas of the fresh catalysts were measured at –196 °C in a JW-BK132F type N₂ physisorption equipment. 200 mg sample was degassed under vacuum at 300 °C for 2 h prior to measurement. The surface area was determined using the Brunauer–Emmett–Teller (BET) equation based on single point method.

The microscopic morphology was observed on a JEOL JSM-6360LV scanning electron microscope (SEM) equipped with a JED-2300 energy dispersive spectroscopy (EDS) surface element analysis system. Sample was prepared using a JEOL JFC-1600 Auto Fine Coater.

Powder XRD measurements for all the fresh and spent catalysts were performed on a Rigaku D/MAX-2200PC X-ray diffractometer with Cu Kα radiation ($\lambda = 1.5406 \text{ \AA}$), 40 kV, 30 mA at scanning speed of 8°/min from 5° (2 θ) to 85°.

H₂ temperature-programmed reduction (H₂-TPR) measurement of the fresh catalysts was carried out on a TP-5000 chemisorption analyzer. Sample of 50 mg was pretreated with He (mL/min) at 350 °C for 1 h, and then cooled to room temperature. The sample was reduced by a reducing gas (25 mL/min) composed of 5% H₂/95% N₂ as the sample was heated from room temperature to 800 °C at a rate of 10 °C/min.

Temperature-programmed desorption (CO₂-TPD) measurement was performed in the same TP-5000 chemisorption analyzer. 200 mg sample was loaded in the reaction tube and was pretreated with H₂/CO (2:1) (30 mL/min) at 300 °C for 4 h, and then purged with He (50 mL/min) at 300 °C for 30 min prior to measurement. Then the sample was adsorbed with CO₂ at 50 °C until saturation, following by purge with He (mL/min) for 30 min. In all the cases, the TPD measurement was performed from room temperature to 700 °C at a heating rate of 10 °C/min.

The X-ray photoelectron spectra (XPS) was recorded on a thermo scientific ESCALAB 250 spectrometer using an Al Kα X-ray source, where the base pressure of the chamber was less than 2×10^{-8} Pa. The binding energies (BEs) were calibrated relative to adventitious carbon using the C 1s peak at 284.8 eV.

2.3. Catalytic activity tests

The CO₂ hydrogenation reaction over prepared Fe–Zn–K catalysts was performed in a stainless flow-type pressurized fixed-bed reactor (8 mm i.d., 400 mm in length) with 2 mL of catalyst. The ratio of height to diameter (H/D) of the catalyst bed was about 5. All the samples were pre-reduced by syngas with H₂/CO of 2 at 1000 h^{–1}, 0.5 MPa and 300 °C for 4 h. H₂ and H₂/CO₂ pretreated 1Fe-1Zn-K sample was also conducted at 1000 h^{–1}, 0.5 MPa and 300 °C for 4 h for comparison. The typical reaction conditions were as follows: H₂/CO₂ = 3 (molar ratio), GHSV = 1000 h^{–1}, $p = 0.5$ MPa and $T = 320$ °C. The effluent gas was analyzed on an online gas chromatograph (GC-9560-I), with a 2 m TDX-01 packed column for C₁ products analysis via TCD detector, and a 50 m Al₂O₃ capillary column for C₁–C₅ hydrocarbons products via FID detector, respectively. The liquid products were analyzed on an off-line gas chromatograph (GC-9560-II), with a 2 m GDX-401 packed column for the aqueous phase products analysis via TCD detector and a 30 m SE-30 capillary column for the oil phase products analysis via FID detector. The mass balance was based on carbon and the results were calculated in terms of activity by CO₂ conversion, light olefin selectivity (wt.) by olefin fraction in overall hydrocarbon products and olefin to paraffin ratio (O/P) in the C₂–C₄ fraction.

3. Results and discussion

3.1. Phase structure of the catalysts

The XRD patterns of the fresh catalyst samples are shown in Fig. 1. The diffraction peaks assigned to rhombohedral hematite (α -Fe₂O₃) [15] were observed for the Fe₂O₃ sample (Fig. 1a) and K promoted Fe₂O₃ sample (Fig. 1b). As the zinc content increased (Fig. 1c–f), the diffraction peaks at $2\theta = 30.5^\circ$, 35.4° , 36.9° , 43.1° ,

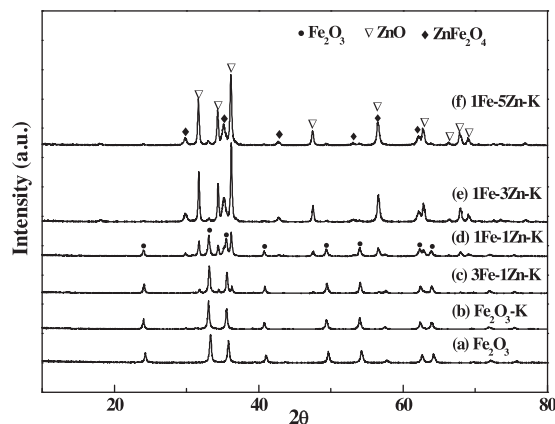


Fig. 1. XRD patterns of fresh catalyst samples.

53.4°, 56.9° and 62.6° appeared, which are assigned to zinc ferrite spinel phase (ZnFe₂O₄) with the franklinite spinel-type structure [20]. In addition, the diffraction peaks assigned to ZnO phase were observed thanks to the addition of zinc in all Zn promoted samples, and the diffraction intensity was gradually increased as the zinc content increased.

For the used catalyst samples, as seen in Fig. 2, Fe₃O₄ phase was found for K promoted Fe₂O₃ sample. With the addition of zinc and increase in the zinc content, most of the iron species were stabilized in the form of ZnFe₂O₄. There were no carbides detected in the bulk after reaction for all the samples. It is known that magnetite phase is responsible for the conversion of CO₂ to CO via the RWGS reaction [21,22] and the surface iron carbide species is active for the hydrogenation of CO to hydrocarbons via FTS [23,24]. However, the surface iron carbides could not be excluded during the in situ reduction of the samples by syngas and during the hydrogenation reaction because of the existence of a large amount of CO produced via the RWGS reaction. No diffractive peaks associated with K for all the fresh and used catalyst samples were detected by XRD due possibly to low concentrations and good dispersion.

3.2. Textural properties of the fresh catalysts

Table 1 shows the textural parameters of the fresh catalysts with different Zn contents by N₂ adsorption. It was found that the surface areas were gradually increased with increasing in the Zn/Fe ratio. The average pore diameter was first increased and then decreased and reached a maximum for 1Fe–1Zn–K catalyst. It is considered that addition of zinc could enhance the dispersion of iron phase and enlarge the surface areas [15,16]. Zinc in ZnFe₂O₄ phase (see Figs. 1 and 2) could also inhibit the sintering of Fe oxides, by keeping iron oxides from sintering during thermal treatments or by providing anchoring or nucleation sites for iron oxides during precipitation or thermal treatments [15], which then improved stability of the catalysts. As shown in Fig. 3, Fe–Zn–K catalyst samples shaped in homogeneous particles within about 100 nm, and could be easily synthesized by the microwave-aided hydrothermal procedure. During this process, the precipitation pH was almost kept constant due to the gradual hydrolysis of urea. This homogeneous precipitation method had advantages in improving the particle uniformity and the dispersion of active components over traditional co-precipitation method. It could be seen that K promoted Fe₂O₃ sample without zinc promotion (Fig. 3a) shaped in regular spheres with homogeneous sizes just as the synthesis of monodispersed nanophase α -Fe₂O₃ by the microwave-hydrothermal method [25]. As the zinc content

increased, the particle sizes were different and irregularly shaped possibly due to the formation of larger ZnO clusters with excessive addition of zinc (Fig. 3c). It was demonstrated that the addition of zinc to iron catalyst has play an important role in modifying structure of the catalysts.

As shown in Fig. 3, Fe–Zn–K catalyst samples shaped in homogeneous particles within about 100 nm, and could be easily synthesized by the microwave-aided hydrothermal procedure. During this process, the precipitation pH was almost kept constant due to the gradual hydrolysis of urea. This homogeneous precipitation method had advantages in improving the particle uniformity and the dispersion of active components over traditional co-precipitation method. It could be seen that K promoted Fe₂O₃ sample without zinc promotion (Fig. 3a) shaped in regular spheres with homogeneous sizes just as the synthesis of monodispersed nanophase α -Fe₂O₃ by the microwave-hydrothermal method [25]. As the zinc content increased, the particle sizes were different and irregularly shaped possibly due to the formation of larger ZnO clusters with excessive addition of zinc (Fig. 3c). It was demonstrated that the addition of zinc to iron catalyst has play an important role in modifying structure of the catalysts.

3.3. Reduction behavior of the catalysts

Fig. 4 shows the TPR profiles of Fe–Zn–K catalysts with varied Fe/Zn molar ratios. The fresh Fe₂O₃ catalyst without zinc promotion (Fig. 4a) appeared two reduction peaks, suggesting a two-step reduction. The first occurred in a sharp reduction peak at low temperature of around 235–410 °C and the other did in a broad reduction peak at high temperature of around 435–700 °C, corresponding to the first reduction of Fe₂O₃ to Fe₃O₄ and then the reduction of Fe₃O₄ to form Fe metal, respectively. The addition of K to Fe₂O₃ (Fig. 4b) caused a slightly shift of the initial reduction peak to higher temperature. Also three peaks appeared, indicating the inhibition effect on the reduction of Fe₂O₃ and the formation of iron phase through FeO [8]. The whole reduction temperatures shifted toward high temperatures with the increase of Zn/Fe ratio (Fig. 4c and d) due to the formation of unreducible ZnFe₂O₄. With further addition of zinc (Fig. 4e and f), the reduction peaks slightly shifted to lower temperatures due to the formation of ZnO clusters, which further improved the dispersion of Fe phase and favored the reduction of iron species.

3.4. Surface adsorption behavior of the catalysts

Fig. 5 shows the CO₂-TPD profiles of Fe–Zn–K catalyst samples. All the samples were pretreated under the similar conditions as the pre-reduction for activity test before CO₂-TPD measurement. The samples were purged with He at 300 °C for 30 min to eliminate absorbed CO species. The TPD profile of the pre-reduced sample Fe₂O₃–K but without CO₂ adsorption (as reference) is shown in Fig. 5(a). It can be seen that the purge caused a remarkable decrease in CO adsorption, and a little of CO adsorption showed almost no effect on the CO₂ adsorption. The CO₂ desorption peaks of Fe₂O₃–K sample located at around 300–420 °C and 450–650 °C, were larger compared with those of zinc promoted samples and the uptake amount was higher than Zn promoted samples (see Table 2). With the addition of zinc, the amount of CO₂ desorption at around 300–420 °C greatly decreased, and the desorption peak at 450–650 °C shifted to lower temperature. A slight increase in CO₂ adsorption on 1Fe–5Zn–K sample compared with 1Fe–3Zn–K sample may be caused by the enhancement of surface basicity on the ZnO clusters [17]. The adsorption behavior of the catalysts is closely related with the phase transformation owing to zinc addition.

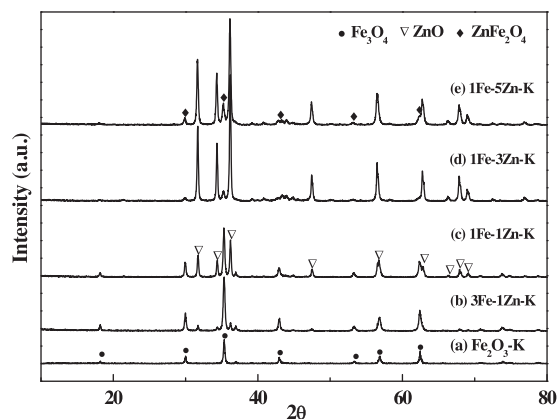
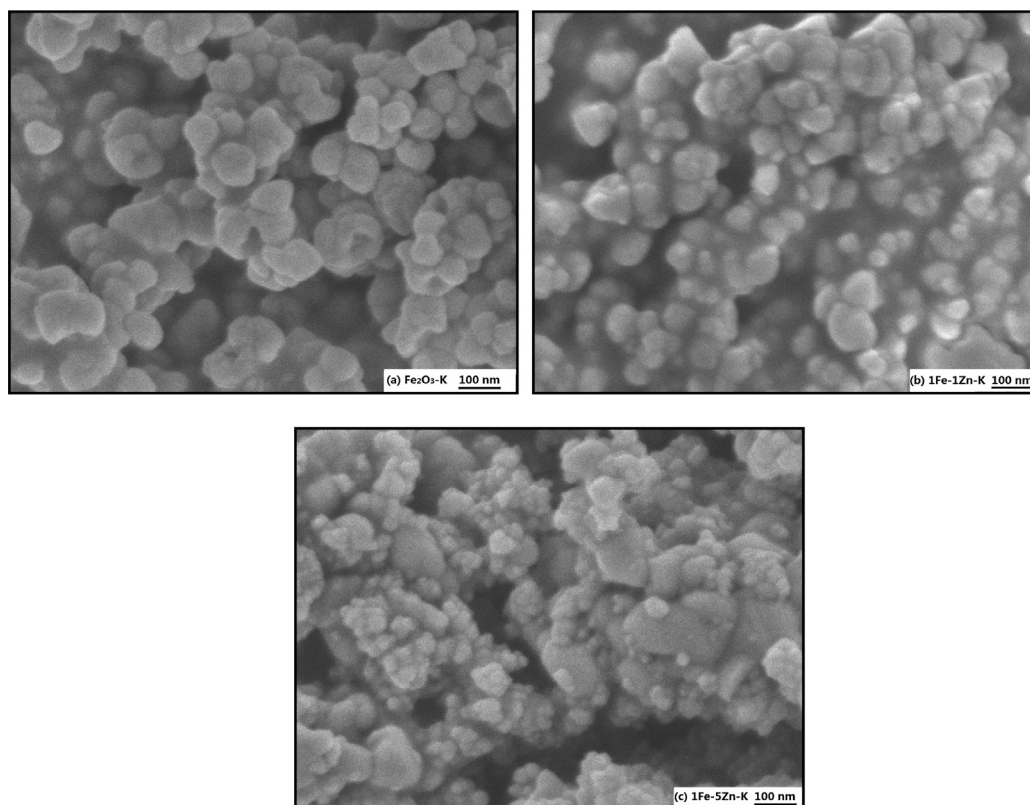
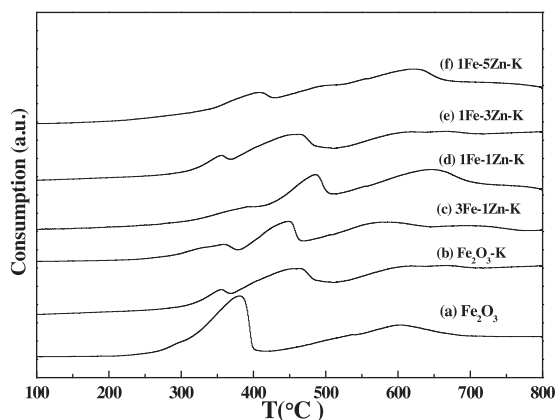


Fig. 2. XRD patterns of catalyst samples after reaction.

Table 1

Textural property of fresh catalyst samples.

Catalyst	BET surface area (m ² /g)	Pore volume (cm ³ /g)	Average pore diameter (nm)
Fe ₂ O ₃ -K	14.40	0.09	25.85
3Fe-1Zn-K	16.35	0.11	25.93
1Fe-1Zn-K	21.17	0.15	27.76
1Fe-3Zn-K	31.47	0.16	20.36
1Fe-5Zn-K	37.72	0.16	16.94

**Fig. 3.** SEM images of catalyst samples.**Fig. 4.** H₂-TPR profiles of catalyst samples.

3.5. XPS measurement of the catalysts

The surface composition of the catalyst samples is shown in Table 3. It was found that the surface molar ratio of Zn and Fe for all the samples was lower than the desired proportion. The K element for all the samples was enriched on the surface of the catalysts and

the Fe/K molar ratio was almost constant except that of 1Fe-5Zn-K sample. The Fe2p XPS spectra are presented in Fig. 6. The binding energy of Fe2p_{3/2} for the sample Fe₂O₃-K without Zn promotion was around 710.6 eV. A distinct satellite peak at a binding energy ~8 eV higher than the main Fe2p_{3/2} peak could be observed. The Fe2p_{3/2} binding energy and peak shape was assigned to characteristic B.E of Fe₂O₃ [26]. The Fe2p spectra peaks shifted to higher binding energies with increased Zn content, suggesting that increased interaction was identified between Fe and Zn with increasing Zn content possibly due to the formation of ZnFe₂O₄, and then changed the chemical environment of Fe.

3.6. Catalytic activity

The activity of different catalyst samples for CO₂ hydrogenation at set reaction conditions is shown in Table 4. It could be seen that ZnO without K promotion sample showed low activity and CH₄ was the main product with 85.32 wt.% in overall hydrocarbon distribution. Fe₂O₃ sample without K promotion also exhibited low C₂=C₄= selectivity and remarkable increase of CH₄ selectivity compared with the samples with both Zn and K promotion. For Fe-Zn-K samples the conversion of CO₂ was above 46% and the C₂-C₄ olefin distribution in overall hydrocarbon products was more than 49 wt.%. The C₅⁺ distribution was less than 5%, except that over

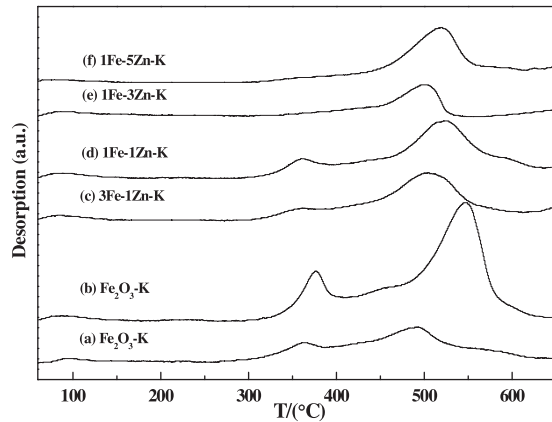


Fig. 5. CO₂-TPD profiles of catalyst samples.
(a) Pretreated by H₂/CO at 300 °C for 4 h without CO₂ adsorption.

Table 2
CO₂-TPD results of different samples.

Samples	Total CO ₂ uptake amount/mmol g ⁻¹
Fe ₂ O ₃ -K	1.501
3Fe-1Zn-K	0.556
1Fe-1Zn-K	0.692
1Fe-3Zn-K	0.278
1Fe-5Zn-K	0.338

Table 3
Surface composition of different samples.^a

Samples	Surface atom content (mol %)						
	Fe	Zn	K	O	C	Fe/Zn	Fe/K
Fe ₂ O ₃ -K	21.91	–	5.60	53.54	18.95	–	3.91
3Fe-1Zn-K	17.38	2.94	4.25	46.73	28.70	5.91	4.09
1Fe-1Zn-K	19.52	5.68	5.00	51.85	17.95	3.44	3.90
1Fe-3Zn-K	19.55	7.01	3.45	47.71	22.28	2.79	5.67
1Fe-5Zn-K	11.99	15.84	3.41	48.55	20.21	0.76	3.52

^a Calculated from the peak area of XPS spectra.

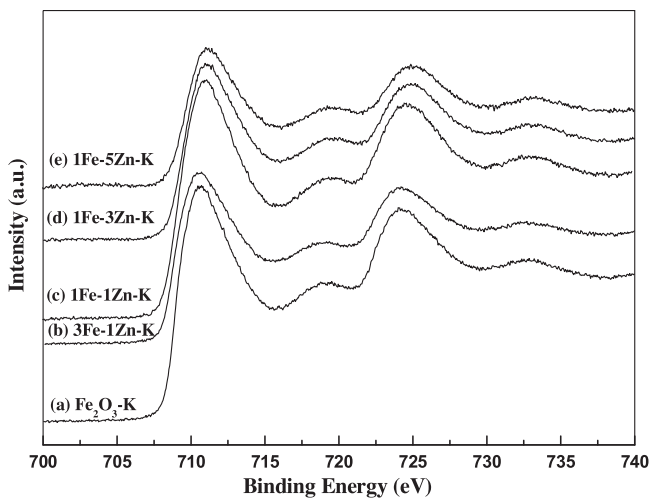


Fig. 6. Fe2p XPS spectra of catalyst samples.

Fe₂O₃-K sample. The CH₄ and CO selectivity was gradually increased and the C₅⁺ selectivity remarkably decreased as the zinc content increased. That is, the addition of zinc suppressed the formation of heavy products and promoted the formation of light

Table 4
Activity of the catalysts for CO₂ hydrogenation.^a

Catalysts	CO ₂ conv.%	Sel. C.% ^c		Hydrocarbon distribution wt.%				
		CH ₄	CO	CH ₄	C ₂ =C ₄ =	C ₂ ⁰ -C ₄ ⁰	C ₅ ⁺	
Fe ₂ O ₃ ^b	44.47	50.87	8.88	58.80	9.49	29.00	2.81	
ZnO ^b	11.08	73.91	12.10	85.32	6.38	8.30	0	
Fe ₂ O ₃ -K ^b	53.94	26.75	5.51	30.67	49.86	6.97	12.50	
3Fe-1Zn-K ^b	52.33	31.56	5.45	36.41	50.65	8.28	4.65	
1Fe-1Zn-K ^b	51.03	29.69	6.00	34.88	53.58	7.81	3.73	
1Fe-3Zn-K ^b	50.48	31.57	7.06	37.26	52.02	7.99	2.73	
1Fe-5Zn-K ^b	46.68	33.40	7.24	39.26	50.16	8.28	2.30	
1Fe-1Zn-K ^c	54.00	26.29	6.29	31.13	50.24	6.46	12.17	
1Fe-1Zn-K ^d	37.76	37.81	11.13	46.32	40.50	7.79	5.40	
1Fe-1Mn-K ^b	46.14	32.57	9.57	39.39	45.41	7.06	8.14	

^a Reaction conditions: H₂/CO₂ = 3/1, GHSV = 1000 h⁻¹, 0.5 MPa, 320 °C, 72 h over time on stream.

^b The samples reduced with H₂/CO (2/1) at 1000 h⁻¹, 0.5 MPa, 300 °C for 4 h.

^c The sample reduced with H₂, 1000 h⁻¹, 0.5 MPa, 300 °C for 4 h.

^d The sample reduced with H₂/CO₂ (3/1), 1000 h⁻¹, 0.5 MPa, 300 °C for 4 h.

^e CH₄ and CO selectivity was calculated based on carbon.

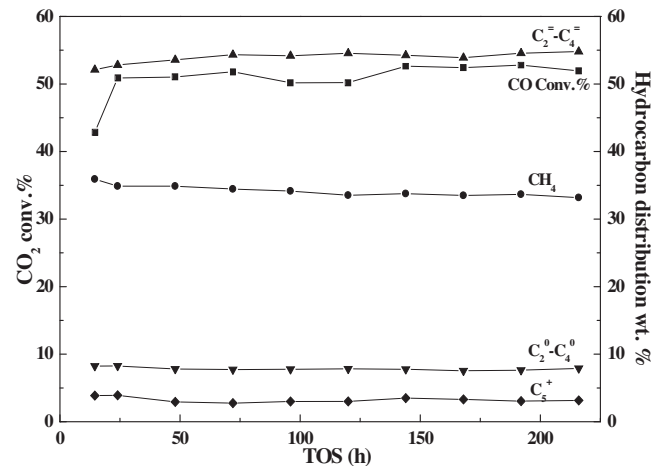


Fig. 7. Activity of 1Fe-1Zn-K catalyst over time-on-stream.
Reaction conditions: H₂/CO₂ = 3, GHSV = 1000 h⁻¹, p = 0.5 MPa and T = 320 °C.

hydrocarbons. A slight increase in the CO and CH₄ formation could be ascribed to the formation of ZnO, which promotes the activity for the RWGS reaction [19] and provides a large amount of activated hydrogen for subsequent hydrogenation of adsorbed reaction intermediates, and thus causes slight increase in the CH₄ selectivity [17]. However, zinc addition has almost no effect on light olefin selectivity, and the olefin/paraffin (O/P) above 6.0 for all the samples, was superior to those by ceria modified Fe/Mn/K catalysts with O/P ratio of 4.4 [27]. These data demonstrate that Fe-Zn-K catalysts have excellent properties toward CO₂ hydrogenation to light olefins. As the Zn content increased, the CO₂ conversion was slightly decreased because of the formation of unreducible ZnFe₂O₄ in iron oxide precursors, which prevented the catalysts from activation and resulted in the activity fall during CO₂ hydrogenation [15]. The 1Fe-1Mn-K sample also showed good activity and olefin selectivity, however, the activity was lower than that of reported FeMnK catalyst with 50.4% CO₂ conversion [27], and the selectivity of C₂=C₄ was lower compared with that of the 1Fe-1Zn-K sample. Pre-reduction with H₂ on 1Fe-1Zn-K catalyst exhibited relatively higher activity and high C₅⁺ selectivity but low CH₄ selectivity in CO₂ hydrogenation, however, increased CH₄ selectivity and low C₂=C₄ selectivity were obtained with H₂/CO₂ reduction. As mentioned above, the prepared Fe-Zn-K catalyst with Fe/Zn molar ratio of 1/1 exhibited the best performance with the CO₂ conversion of 51.03% and the C₂-C₄

olefins of 53.58 wt.% in overall hydrocarbons. Although the CO₂-TPD results showed that CO₂ adsorption was greatly inhibited on Zn promoted samples but did not result in a significant decrease in activity, suggesting that the FT reaction becomes the rate-controlling step in overall conversion of CO₂ to olefins, and Fischer–Tropsch (FT) reaction rate is much slower than the reverse water gas shift (RWGS) reaction rate [28]. Also, with the increase of Zn content, only ZnO and ZnFe₂O₄ phases were observed without single iron species for both fresh and used samples. The sample 1Fe–5Zn–K with unreducible ZnFe₂O₄ also exhibited high activity in CO₂ hydrogenation, suggesting that the synergistic effect between ZnO and ZnFe₂O₄ resulted in improving the activity of CO₂ hydrogenation.

The activity of 1Fe–1Zn–K catalyst sample with time on stream for CO₂ hydrogenation is shown in Fig. 7. It is clear that the catalyst showed excellent stability along with the 216 h reaction, due possibly to the presence of ZnFe₂O₄ phase. This zinc phase proves to be able to stabilize the surface active sites for FTS [16]. High CO₂ conversion of 42.85% at the initial stage of reaction was obtained by the syngas pretreatment with H₂/CO (2:1 molar ratio), implying that the catalyst surface was activated for both the RWGS reaction and the CO hydrogenation to hydrocarbons. It could also be seen that the content of C₂–C₄ olefin products exceeded 51 wt.% in overall hydrocarbon distribution, higher than those by K promoted Fe/ZrO₂ catalyst with 46% fraction of C₂–C₄ olefins in CO₂ hydrogenation [3]. Based on the above activity behaviors and the structural characters of the as-prepared Fe–Zn–K catalysts, it was concluded that the current Zn promoted iron catalysts are favorable for CO₂ hydrogenation to light olefins with high activity and stability.

4. Conclusions

To improve the selectivity of light olefin in CO₂ hydrogenation, Fe–Zn–K catalysts were prepared by microwave-aided hydrothermal procedure and subsequent potassium promotion by incipient wetness impregnation. All the catalyst samples displayed good activity and light olefin selectivity in CO₂ hydrogenation. K promoted iron catalyst showed high initial activity and was more likely to be oxidized to Fe₃O₄ during the reaction. With the addition of zinc, the C₅⁺ products were remarkably suppressed. The selectivity of CH₄ and CO was slightly increased as further addition of zinc. The stability of the catalyst was improved owing to the formation of ZnFe₂O₄ phase. The addition of zinc played an important role in tailoring CO₂ hydrogenation activity and product distribution by modifying the structure and surface adsorption

behavior of iron catalysts, and thus improving the product distribution and stability of the catalysts.

Acknowledgments

The authors gratefully acknowledged the financial supports from the National Natural Science Foundation of China (21366025), the Western Light Talent Culture Project of Chinese Academy of Sciences and the prophase research special project of 973 (2012CB723106) from the Chinese Ministry of Science & Technology.

References

- [1] M. Martinelli, C.G. Visconti, L. Lietti, P. Forzatti, C. Bassano, P. Deiana, *Catal. Today* 228 (2014) 77–88.
- [2] M. Mikkelsen, M. Jørgensen, F.C. Krebs, *Energy Environ. Sci.* 3 (2010) 43–81.
- [3] J. Wang, Z. You, Q. Zhang, W. Deng, Y. Wang, *Catal. Today* 215 (2013) 186–193.
- [4] S. Saeidi, N.A. Saidina Amin, M.R. Rahimpour, *J. CO₂ Util.* 5 (2014) 66–81.
- [5] S. Kiatphuegpon, M. Chareonpanich, J. Limtrakul, *Chem. Eng. J.* 240 (2014) 527–533.
- [6] M.L. Cubeiro, H. Morales, M.R. Goldwasser, M.J. Pérez-Zurita, F. González-Jiménez, C. Urbina de N, *Appl. Catal. A Gen.* 189 (1999) 87–97.
- [7] M.L. Cubeiro, H. Morales, M.R. Goldwasser, M.J. Pérez-Zurita, F. González-Jiménez, *React. Kinet. Catal. Lett.* 69 (2000) 259–264.
- [8] S.S. Nam, H. Kim, G. Kishan, M.J. Choi, K.W. Lee, *Appl. Catal. A Gen.* 179 (1999) 155–163.
- [9] H. Schulz, *Appl. Catal. A Gen.* 186 (1999) 3–12.
- [10] C.F. Huo, B.S. Wu, P. Gao, Y. Yang, Y.W. Li, H.J. Jiao, *Angew. Chem. Int. Ed.* 50 (2011) 7403–7406.
- [11] H.M. Torres Galvis, K.P. de Jong, *ACS Catal.* 3 (2013) 2130–2149.
- [12] R. Saththawong, N. Koizumi, C. Song, P. Prasassarakich, *J. CO₂ Util.* 3–4 (2013) 102–106.
- [13] J.F. Lee, W.S. Chern, M.D. Lee, T.Y. Dong, *Can. J. Chem. Eng.* 70 (1992) 511–515.
- [14] J.S. Kim, S. Lee, S.B. Lee, M.J. Choi, K.W. Lee, *Catal. Today* 115 (2006) 228–234.
- [15] S. Li, A. Li, S. Krishnamoorthy, E. Iglesia, *Catal. Lett.* 77 (2001) 197–205.
- [16] H. Wang, Y. Yang, J. Xu, H. Wang, M. Ding, Y. Li, *J. Mol. Catal. A Chem.* 326 (2010) 29–40.
- [17] X.M. Liu, G.Q. Lu, Z.F. Yan, J. Beltramini, *Ind. Eng. Chem. Res.* 42 (2003) 6518–6530.
- [18] T. Shido, Y. Iwasawa, *J. Catal.* 129 (1991) 343–355.
- [19] M. Saito, J. Wu, K. Tomoda, I. Takahara, K. Murata, *Catal. Lett.* 83 (2002) 1–4.
- [20] D. Wang, X. Zhang, Y. Gao, F. Xiao, W. Wei, Y. Sun, *Catal. Commun.* 11 (2010) 430–433.
- [21] T. Riedel, H. Schulz, G. Schaub, K.W. Jun, K.W. Hwang, J.S. Lee, *Top. Catal.* 26 (2003) 41–54.
- [22] E.S. Lox, G.F. Froment, *Ind. Eng. Chem. Res.* 32 (1993) 71–82.
- [23] T. Herranz, S. Rojas, F.J. Pérez-Alonso, M. Ojeda, P. Terreros, J.L.G. Fierro, *J. Catal.* 243 (2006) 199–211.
- [24] Q. Zhang, W. Deng, Y. Wang, *J. Energ. Chem.* 22 (2013) 27–38.
- [25] H. Katsuki, S. Komarneni, *J. Am. Ceram. Soc.* 84 (2001) 2313–2317.
- [26] J. Zhang, L. Ma, S. Fan, T. Zhao, Y. Sun, *Fuel* 109 (2013) 116–123.
- [27] R.W. Dörner, D.R. Hardy, F.W. Williams, H.D. Willauer, *Catal. Commun.* 15 (2011) 88–92.
- [28] H.D. Willauer, R. Ananth, M.T. Olsen, D.M. Drab, D.R. Hardy, F.W. Williams, *J. CO₂ Util.* 3–4 (2013) 56–64.